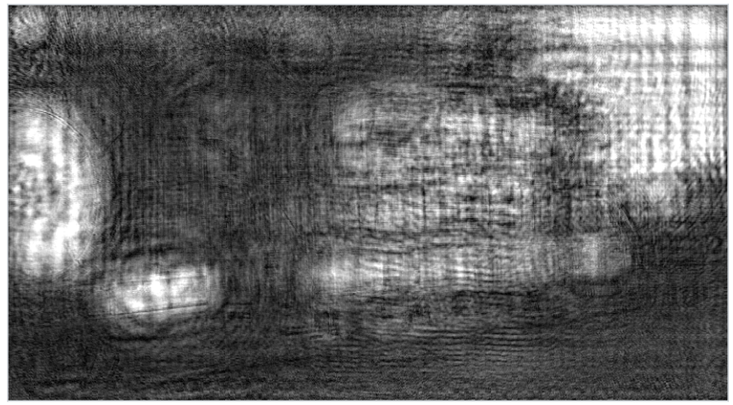
One phase-only hologram, two reconstruction depths

18 cm / 22 cm | Frozen initial-phase and final encoding networks | No additional training

Desired content / target
18 cm layer mask



Without carrier alignment
13.61 dB (assigned region)



With carrier alignment
25.92 dB (assigned region)

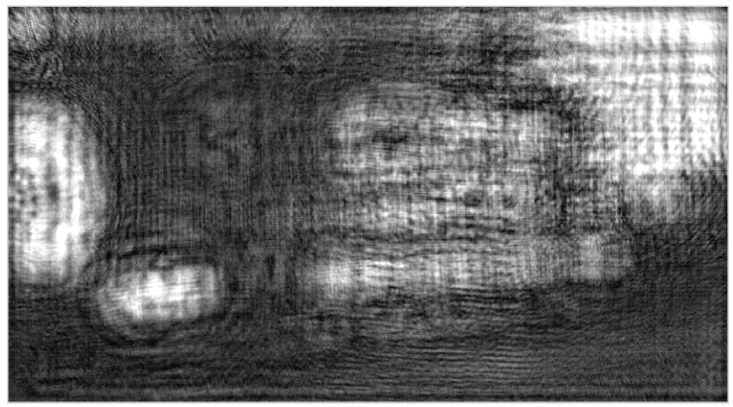


18 cm plane

22 cm layer mask



14.17 dB (assigned region)



24.58 dB (assigned region)

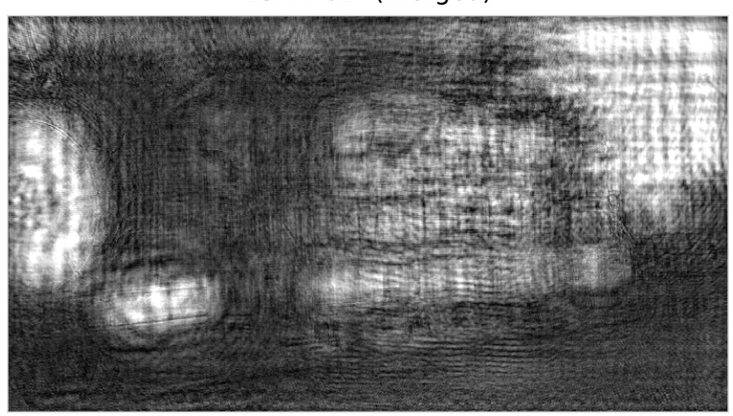


22 cm plane

Full target



13.77 dB (merged)



25.47 dB (merged)



Depth-masked merge

Each reconstruction column comes from ONE hologram; the last row selects pixels by the depth mask.
Black areas in the target layer mark unscored pixels, not required darkness. Defocused light remains visible in the complete planes.
One fitted brightness factor is shared by both planes and the merge within each method.